Machine Learning Reveals Welfare Discourse Embedded in Japanese Sentencing: NLP Analysis of 2,002 Theft Judgments

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1.85×

pooled IG attribution magnitude of lexicon-matched vs. other WordPieces (BERT's subword tokens; contrast not matched on frequency or part of speech; judgment-cluster 95% CI [1.60, 2.10]; fold range 1.50–2.36×)

9 / 59

lexicon features kept by the sparse L1 model: 8 toward suspension, 1 toward custody (*shakai fukki*, social reintegration)

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lexicon features in the top 10% of the 444 features the sparse L1 model keeps: 3.0× a frequency-matched expectation (1.0), p = 0.055, unstable across folds

0.947

BERT AUC (10-fold mean), custodial vs. suspended from text alone (linear 0.900 with prior convictions; 0.901 text-only)

1 · Introduction

- Japan has no pre-sentence investigation (PSI) system for adult criminal courts — welfare information reaches judges mainly through ad hoc advocacy, not systematic assessment.
- In the focal-concerns account, sentencers weigh blameworthiness, community protection, and practical constraints (Steffensmeier, Ulmer & Kramer 1998), and recent work adds the defendant's redeemability (Ulmer, Silver & Hanrath 2023); public beliefs about redeemability predict punitiveness (Maruna & King 2009).
- Question: how does welfare-related language in sentencing reasoning relate to custody when no PSI system exists?
- Japanese legal NLP has analysed citation structure and administrative reasoning paths (Kondo et al. 2023; 2025); sentencing reasons themselves have been examined with text mining (Kanda 2013), and judgment texts and news reports have been mined for factors behind sentencing trends (Rzepka et al. 2008); we add an explainable-ML audit of welfare vocabulary with frequency-matched nulls and judgment-clustered inference. The model is a discourse-analysis instrument — not a sentencing tool.

2 · Data & Provenance

- 1,954 unique criminal judgments retrieved from Westlaw Japan by full-text theft-sentencing queries; the 2,002 rows of the analysis table behind the accepted title included 48 duplicate retrievals, removed before every result shown here.
- Outcome labels: custodial 1,593 (81.5%) vs. suspended 361 (18.5%).
- Curated legal databases over-represent contested, legally noteworthy cases — findings describe the world of published, contested sentencing, not routine practice.

DATA-PROVENANCE NOTE (clarifying the submitted abstract)

The corpus was retrieved from Westlaw Japan with full-text theft-sentencing queries. On auditing the bibliographic metadata this set proved heterogeneous — 16.6% are identified as theft-principal cases, judgment years span 1958–2026 (not a nominal 2010–2023), and 3.5% are appellate. The results shown here come from a re-run on the 1,954 unique judgments with the linear model re-estimated (linear AUC 0.900 with prior convictions; BERT AUC 0.947, 10-fold mean); within the theft-principal subset BERT's discrimination holds (0.945, pooled) while the linear model's falls (0.884 with prior convictions; 0.854 text only); the lexicon analyses use the full corpus.

3 · Methods

- Sentencing-reasons sections extracted from each judgment → tokenized with MeCab (UniDic dictionary); lines that restate the operative sentence or the applied statutes are removed, and the TF-IDF pipeline also removes nine disposition and statute lemmas.
- TF-IDF representation (5,000 features) + one structured field (prior-conviction count) = 5,001 features.
- L1-regularized logistic regression, 10-fold stratified cross-validation → exact SHAP values (importance = mean |SHAP|; direction = sign of the fitted weight); the text-only variant (5,000 features) is the linear model in Figure 1, while the lexicon analyses use the 5,001-feature model with prior convictions (444 features kept).
- Fine-tuned Japanese BERT (tohoku-nlp/bert-base-japanese-v3; main run with a fixed seed (42); fold-mean AUC 0.939–0.947 over 4 seeds) → an Integrated Gradients (IG) audit on folds 1–3 of the ten validation folds.
- A lexicon of 59 welfare-support features, expanded from 31 stem entries drawn from Japanese scholarship on welfare in sentencing (the stems play no part in fitting), evaluated against the importance rankings (enrichment = more lexicon features in the top 10% of the features the model keeps than non-welfare vocabulary yields when drawn from the kept features and matched on document frequency and n-gram order).
- Secondary analysis of published judgments; defendants' names redacted at source.

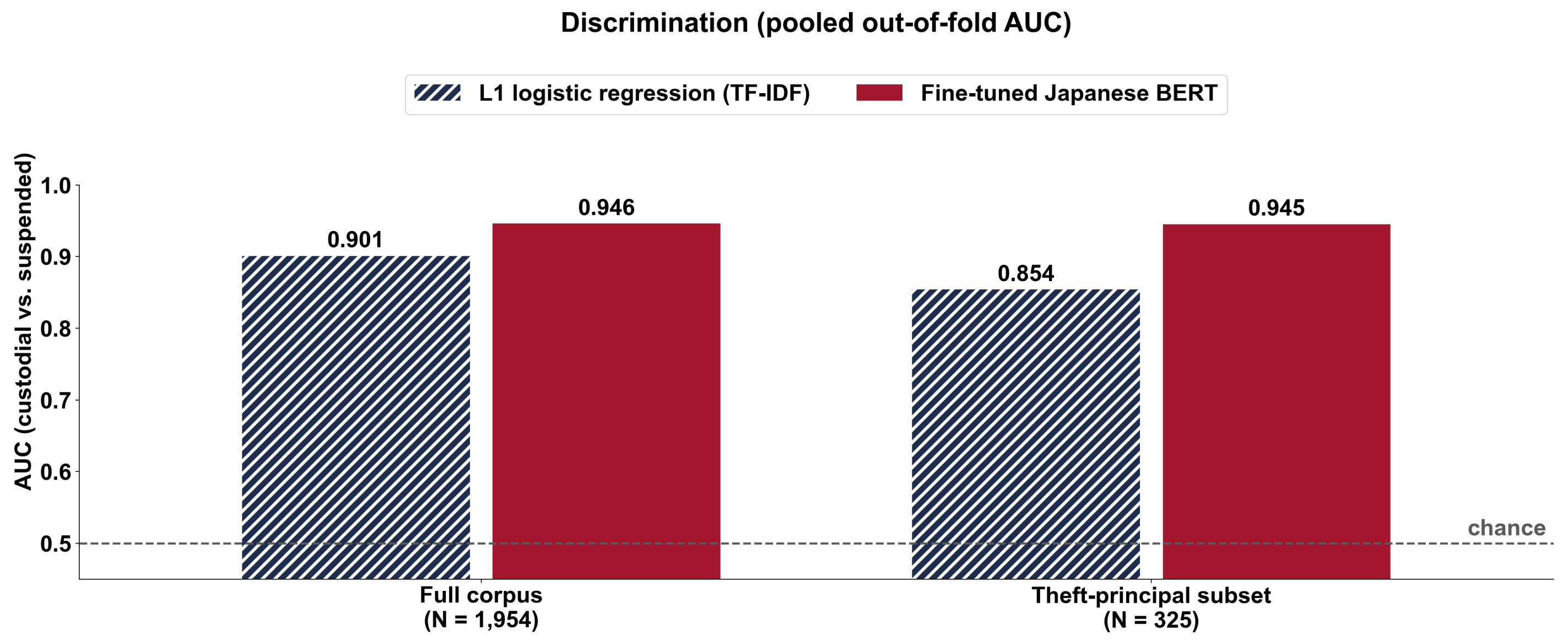

Figure 1. Discrimination (pooled out-of-fold AUC, custodial vs. suspended) on the full corpus and within the theft-principal subset — BERT's discrimination is unchanged; the text-only linear result is more sensitive to composition (linear model with prior convictions added: 0.900 / 0.884).

4 · Results

- In the BERT audit, lexicon-matched tokens (about half of them bigram-completing morphemes, counted wherever they occur) attract 1.50–2.36× the mean absolute attribution of all other WordPieces, function words included (the contrast is matched neither on frequency nor on part of speech), with judgment-level cluster bootstrap intervals excluding parity in every fold (pooled 1.85×, 95% CI [1.60, 2.10]), although the per-judgment median ratio is smaller (1.18 pooled) and excludes parity only in one fold and pooled.
- In the sparse linear model, lexicon enrichment is weak and fragile. The L1 penalty keeps 9 of the 59 lexicon features, 8 toward suspension (for example support, a pledge to supervise); the one custody-weighted feature, *shakai fukki* (social reintegration), is a bigram that outweighs the contribution of the suspension-weighted unigram it contains (toward custody in 224 of the 232 judgments that contain it).
- Three of the 59 lexicon features rank in the top 10% of the 444 features the model keeps (3.0× a frequency-matched expectation, p = 0.055; p from 0.002 to 0.57 across folds).
- The outcome is largely recoverable from the written reasons — linear AUC = 0.90 (0.901 from text alone; 0.900 with prior convictions); BERT AUC = 0.947 (10-fold mean; pooled 0.946, Figure 1).
- Prior-conviction count ranks third (mean |SHAP| = 0.356, toward custody); adding it does not change full-corpus AUC (−0.002, 95% CI [−0.014, 0.010]).
- Within the theft-principal subset (N = 325): AUC 0.854 (linear, text only) / 0.945 (BERT, pooled).

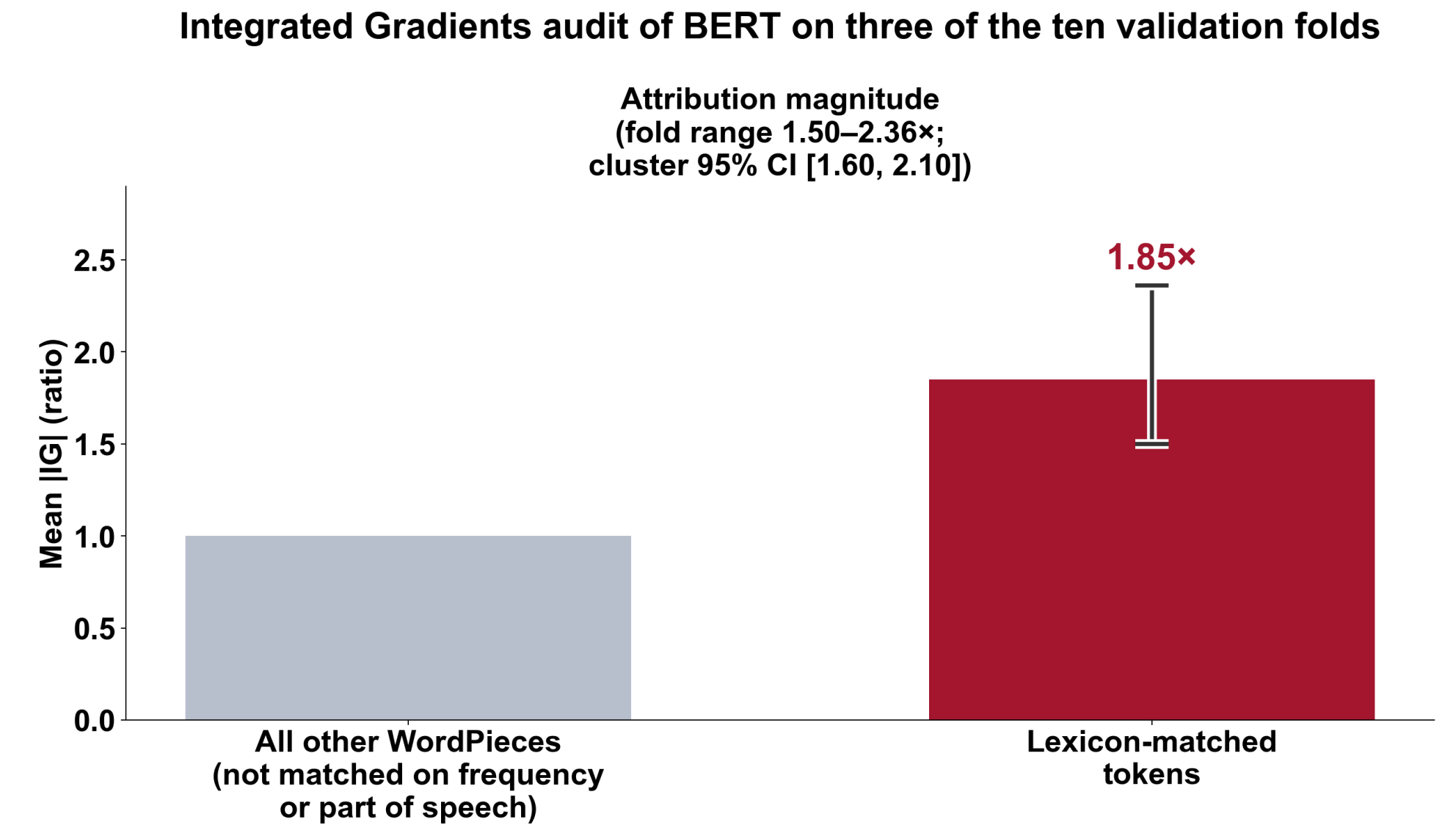

Figure 2. IG audit of the fine-tuned BERT on three of the ten validation folds: pooled ratio of mean absolute attribution, lexicon-matched tokens vs. all other WordPieces (error bar: range across the three folds); the contrast is not compared with a frequency-matched null.

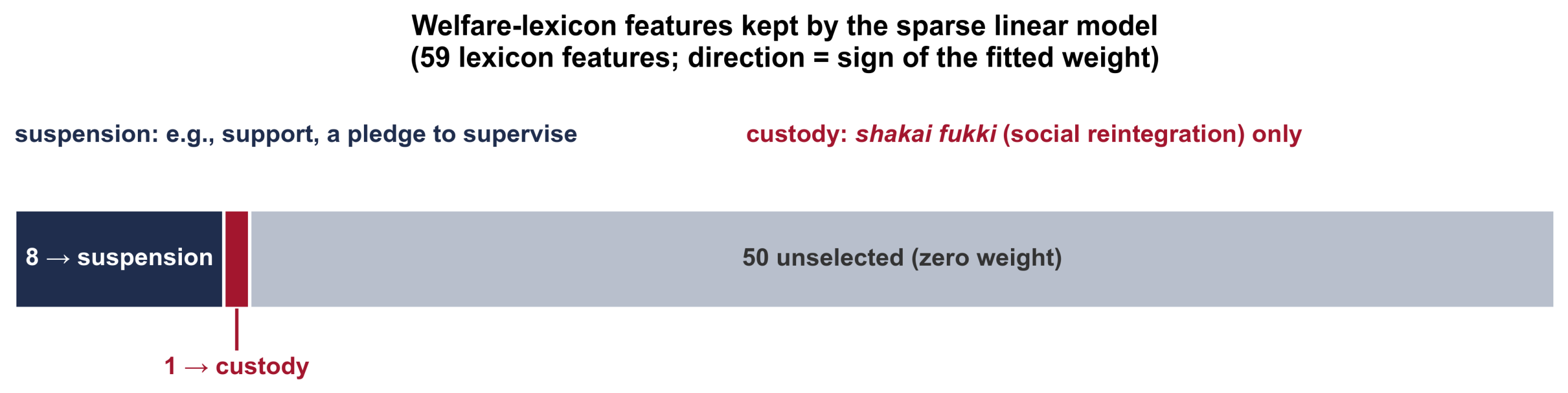

Figure 3. Direction of the fitted weight across the 59 lexicon features:
8 toward suspension, 1 toward custody, 50 unselected.
Of the 9 kept features, 3 rank in the top 10% of the features the model keeps (p = 0.055).

5 · Discussion & Policy

- Welfare vocabulary attracts attribution in the language model, while its enrichment in a sparse linear model is weak and fragile (the two tests differ: the attribution ratio has no frequency-matched null, whereas the linear enrichment is tested against one).
- For policy the consequence is a question, not a recommendation: who controls the supply of welfare information.
- A judgment records only the court's rendering of what was submitted; a companion study by the authors (manuscript under review) coded the support information recorded in a re-collected sample of theft judgments and examined its association with custody.
- If which welfare circumstances reach the record depends on defense resources, a model fit to this record could, if deployed, reproduce at scale whatever unequal access to legally effective narrative the record already reflects.
- Transferability: The audit design (matched nulls, cluster inference, cross-fold stability grading) can be re-implemented elsewhere; the tokenization, extraction rules, and lexicon are specific to Japanese judgments and this offense domain, so any transfer requires re-derivation and hand validation of the vocabulary.

Cited: Steffensmeier, Ulmer & Kramer (1998) Criminology 36(4) · Ulmer, Silver & Hanrath (2023) Justice Quarterly 40(6) · Maruna & King (2009) Eur. J. Crim. Policy Res. 15 · Kondo, Yoshida & Hisano (2023) Artif. Intell. Law 31 · Kondo et al. (2025) K-CAP · Kanda (2013) Kinki Daigaku Hōgaku 61(2–3) · Rzepka et al. (2008) NTCIR-7.

TAKE-HOME

Without a pre-sentence investigation system, welfare vocabulary in these published Japanese sentencing reasons attracts attribution in BERT, while its enrichment in a sparse linear model is weak and fragile.

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Poster Session I · Thursday, November 19, 2026 · 6:30–7:30 pm

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